Biospectra : ISSN: 0973-7057, Vol. 19(1), March, 2024, pp. 113-116
An International Biannual Refereed Journal of Life Sciences
Animal Science

ISSN : 0973-7057

Groundwater as a vital resource: A physicochemical study of water quality from different sources of Supaul, Bihar

Kumari Simmi & Arun Kumar*
University Department of Zoology, B. N. Mandal University, Madhepura, Bihar, India

Received : 24th January, 2024 ; Revised : 25th February, 2024
DOI:-https://doi.org/10.5281/zenodo.15048709

Abstract- Groundwater is a crucial natural resource that supports drinking water supply, agriculture, and industrial activities. This study evaluates the physicochemical characteristics of groundwater collected from ten distinct bore wells and open wells. Parameters such as temperature, pH, total dissolved solids (TDS), dissolved oxygen (DO), dissolved carbon dioxide (CO_2), alkalinity, magnesium content, total hardness, and sulfate concentration were analyzed over four months using standard laboratory techniques. The results indicate that the water samples exhibit a near-neutral pH (6.0-7.1), moderate TDS levels (150-201 ppm), and acceptable dissolved oxygen concentrations (6.0-6.2 ppm). Bore well water samples generally exhibited lower TDS but slightly higher dissolved CO_2 levels compared to open well samples. Open well samples displayed higher alkalinity (158-295 ppm) and total hardness (226-502 ppm), possibly due to surface runoff and prolonged atmospheric exposure. While most parameters fall within permissible limits for potable water, certain samples exhibited high hardness levels, indicating potential treatment requirements before consumption. The study emphasizes the need for regular monitoring and sustainable groundwater management practices to ensure long-term water quality and availability.

Key words: pH, TDS, DO, CO_2, groundwater, alkalinity

INTRODUCTION

Groundwater is one of the most critical natural resources on Earth, serving as a lifeline for millions of people and sustaining agricultural productivity. It is stored beneath the Earth's surface in aquifers-layers of rock and soil that hold water.[1] This hidden reservoir plays a crucial role in maintaining ecosystems, supporting livelihoods, and ensuring water security for present and future generations. Groundwater is the primary source of drinking water for nearly half of the global population.[2,3] In rural areas, where surface water sources may be scarce or contaminated, wells and boreholes provide safe and reliable drinking water. Many cities depend on groundwater for municipal water supply, ensuring a continuous flow of clean water for domestic use. Industries, including manufacturing, food processing, and pharmaceuticals, rely heavily on groundwater for production processes. It is used for cooling machinery, generating power, and maintaining industrial operations, contributing to economic growth.[4-6] Groundwater supports ecosystems by maintaining base flow in rivers, lakes, and wetlands. It plays a key role in regulating temperature and sustaining aquatic habitats, benefiting biodiversity. Agriculture is the largest consumer of groundwater, accounting for nearly 70% of its use worldwide. Farmers depend on wells and tube wells to irrigate crops, especially in regions with limited rainfall. Reliable access to groundwater ensures consistent food production, reducing

*Corresponding author :
Phone : 9006991000
E-mail : prf.arunkumar@gmail.com

dependence on erratic monsoons and droughts. Groundwater serves as a buffer during dry seasons, helping farmers sustain crops when surface water sources dry up. In arid and semi-arid regions, groundwater is often the only dependable water source for agriculture.[6,7] Groundwater is essential for providing drinking water to livestock, which is a significant part of rural economies. Many small-scale farmers and pastoral communities rely on groundwater-fed wells for their survival. Excessive withdrawal of groundwater for irrigation and urbanization leads to declining water tables, making it difficult to access water in the future.[8] Pollution from industrial waste, agricultural runoff (pesticides and fertilizers), and improper waste disposal threatens groundwater quality. Adopting rainwater harvesting, efficient irrigation techniques, and groundwater recharge programs can help sustain this precious resource. Numerous pollutants have an impact on water quality. Because of the activity of the microorganisms present, drinking contaminated water might lead to major health issues. In addition to causing dangerous diseases, the action of microorganisms degrades water quality. Therefore, the purpose of this study is to use physicochemical investigations to evaluate the quality of water from various sources.

MATERIALS & METHODS

For the project, water samples were taken from ten distinct tube wells and wells of Kishanpur area of Supaul district. A closed bottle was used to collect the sample. Chemical and biological processes may alter the water sample's quality from the moment of collection. We must preserve as soon as the water sample is collected in order to reduce the growth of microorganisms. Following the collection of the water sample, the TDS, taste, and odor should be assessed and preserved by reducing the temperature and applying chemical preservations. Four months were dedicated to the water analysis process. After the water samples were obtained, they were taken to the lab for analysis. A pH meter was used to measure the pH, and a turbidity meter is used to measure the turbidity.

RESULTS & DISCUSSION

Physicochemical Properties of Water Samples

Temperature

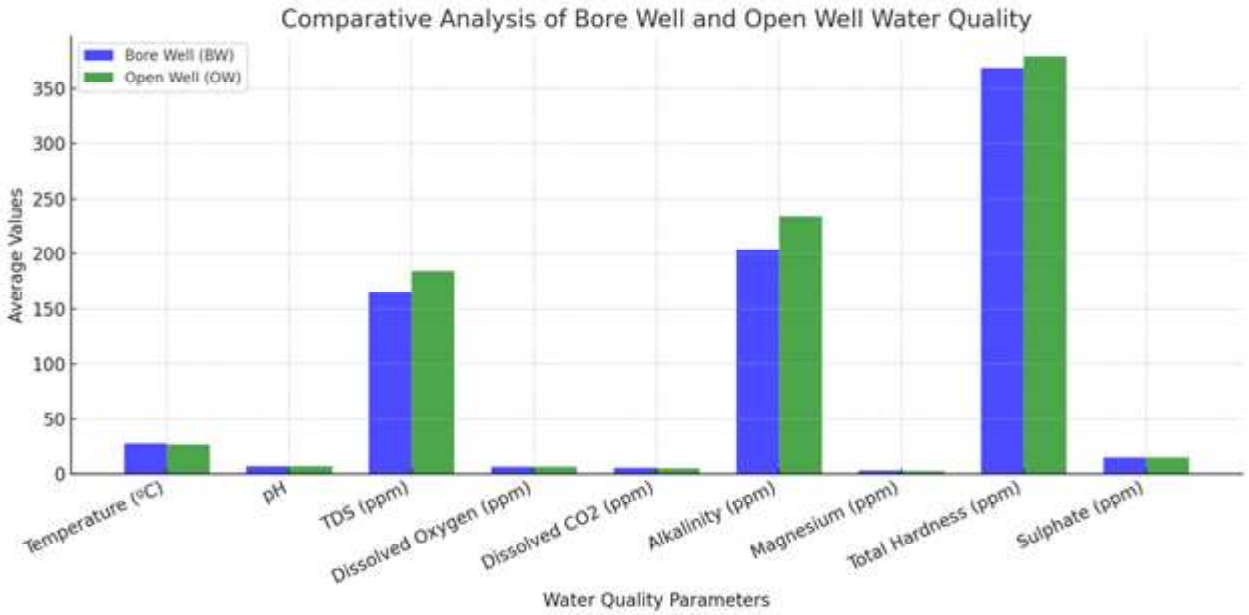
The temperature of the collected water samples varied between 26ºC and 29ºC. The bore well (BW) water samples exhibited slightly higher temperatures (27ºC to 29ºC) compared to the open well (OW) samples (26ºC to

Table 1- Physicochemical parameters of collected ground water samples of Bore wells

Sl. No.	Test	BW1	BW2	BW3	BW4	BW5
1	Temperature (ºC)	27	29	28	27	28
2	pH	6.8	7.1	6.9	6.9	7.0
3	TDS (ppm)	150	155	160	184	175
4	Dissolved Oxygen (ppm)	6.0	6.1	6.2	6.2	6.1
5	Dissolved Carbon Dioxide (ppm)	5.7	5.9	4.4	5.0	5.7
6	Alkalinity (ppm)	201	148	250	208	210
7	Magnesium (ppm)	2.2	3	3	3.5	2.5
8	Total Hardness (ppm)	280	305	346	501	410
9	Sulphate (ppm)	11	15	18	14	19

Table 2- Physicochemical parameters of collected ground water samples of Open wells

Sl. No.	Test	OW1	OW2	OW3	OW4	OW5
1	Temperature (ºC)	26	28	27	27	27
2	pH	6.5	7.0	6.9	6.0	7.1
3	TDS (ppm)	201	188	175	181	177
4	Dissolved Oxygen (ppm)	6.1	6.2	6.2	6.1	6.0
5	Dissolved Carbon Dioxide (ppm)	5.2	5.1	4.2	5.1	5.4
6	Alkalinity (ppm)	255	158	218	295	241
7	Magnesium (ppm)	2.0	2.1	1.9	3.9	2.8
8	Total Hardness (ppm)	226	396	355	416	502
9	Sulphate (ppm)	17	14	16	11	18



Graph 1- Comparative analysis of physicochemical parameters of collected ground water samples of Open wells and Bore wells

28ºC). These values are within an acceptable range for groundwater and indicate minimal thermal pollution.

pH

The pH values of the water samples ranged from 6.0 to 7.1. The bore well samples had a pH range of 6.8 to 7.1, while the open well samples ranged from 6.0 to 7.1. Most samples were near neutral, but OW4 had a slightly acidic nature (pH 6.0), which may indicate the presence of dissolved carbon dioxide or organic matter. Generally, the pH values comply with the acceptable range for potable water (6.5 to 8.5).

Total Dissolved Solids (TDS)

TDS values varied between 150 ppm and 201 ppm. Bore well samples exhibited a lower range (150-184 ppm), whereas open well samples had slightly higher TDS values (175-201 ppm). Higher TDS in OW samples suggests greater mineral dissolution due to prolonged exposure to environmental factors.

Dissolved Oxygen (DO)

The DO levels in the samples ranged from 6.0 ppm to 6.2 ppm. Open well samples showed slightly higher values compared to bore well samples, which could be due to greater exposure to atmospheric oxygen. The DO levels are within the acceptable limit for drinking water quality and indicate good aeration.

Dissolved Carbon Dioxide (CO_2)

The dissolved CO_2 concentrations varied between 4.2 ppm and 5.9 ppm. Bore well samples exhibited slightly higher CO_2 content, possibly due to underground microbial activity and limited gaseous exchange. The lowest value was observed in OW3 (4.2 ppm), indicating lesser organic decomposition.

Alkalinity

Alkalinity values ranged from 148 ppm to 295 ppm. Bore well samples exhibited moderate alkalinity (148-250 ppm), while open well samples had a wider range (158-295 ppm), with OW4 showing the highest value (295 ppm). Higher alkalinity in open wells may be attributed to surface runoff carrying dissolved bicarbonates and carbonates.

Magnesium Content

Magnesium concentration varied between 1.9 ppm and 3.9 ppm. OW3 had the lowest value (1.9 ppm), whereas OW4 had the highest (3.9 ppm). Bore well samples ranged from 2.2 ppm to 3.5 ppm. These levels are within the permissible limits but contribute to total hardness.

Total Hardness

The total hardness of the samples ranged from 226 ppm to 502 ppm. Bore well samples exhibited moderate to high hardness (280-501 ppm), whereas open well samples showed a wider variation, with OW5 having the highest value (502 ppm). Elevated hardness levels indicate

a higher concentration of dissolved calcium and magnesium, making the water less suitable for domestic use.

Sulphate Content

Sulphate levels in the water samples ranged from 11 ppm to 19 ppm. Bore well samples showed values between 11-19 ppm, while open well samples had a range of 11-18 ppm. These levels are well below the permissible limit, indicating no significant sulfate contamination.

Overall, the study revealed that bore well and open well water samples had similar physicochemical characteristics, with some variations. Open well samples exhibited slightly higher TDS, alkalinity, and total hardness, indicating possible environmental influences such as surface runoff and exposure to atmospheric gases. Bore well samples showed slightly higher dissolved carbon dioxide levels, likely due to underground microbial activity.

CONCLUSION

The results suggest that both bore well and open well water sources are generally suitable for drinking and domestic use. However, certain parameters, such as total hardness and alkalinity in some samples, may require treatment before consumption. Continuous monitoring and water quality management practices are recommended to ensure safe water usage.

ACKNOWLEDGEMENT

The authors are thankful to the faculties and researchers of University Department of Zoology, B.N.M.University, Madhepura, Bihar for their constant support and encouragement during the research work.

REFERENCES

1. **APHA. 1985.** Standard Methods for Examination of Water and Wastewater, 20th Edition, American Public Health Association, Washington D.C.
2. **Ravikumar, P., Somashekar, R. K., & Angami, M. 2011.** Assessment of groundwater quality for drinking purpose in rural areas surrounding a defunct copper mine. *Aquatic Procedia,* **4:**685-692.
3. **Jin Hur , Bo-Mi Lee , Tae-Hwan Lee and Dae-Hee Park. 2010.** Estimation of Biological Oxygen Demand and Chemical Oxygen Demand for Combined Sewer Systems Using Synchronous Fluorescence Spectra; *Sensors*, **10:**2460-2471.
4. **Kumar A., Singh A. K., & Ramanathan A. L. 2018.** Groundwater quality assessment in northwestern Bihar alluvial aquifer, India: A geochemical approach. *Groundwater for Sustainable Development,* **6:** 42-54.
5. **Bartram J., K. Lewis, R. Lenton and A. Wright. 2005.** Focusing on improved water and sanitation for health. *Lancet*, **365(9461):** 810-812
6. **Basavaraja Simpi, S. M. Hiremath, K. N. S. Murthy, K. N. Chandrashekarappa, Anil N. Patel, E. T. Puttiah. 2011.** Analysis of Water Quality Using Physico-Chemical Parameters Hosahalli Tank in Shimoga District, Karnataka, India. *Global Journal of Science Frontier Research,* **1(3):** 31-34.
7. **Barbara, A.D. and I.N. Harry. 1984.** Relationship between Ground Water Silica, Total Dissolved Solids and Specific Electric Conductivity. *Ground Water,* **22(1):** 80-85.
8. **Selvam S., Manimaran G., Sivasubramanian P., Balasubramanian N., & Seshunarayana T. 2014.** GIS-based evaluation of water quality index of groundwater resources around Tuticorin coastal city, South India. *Environmental Earth Sciences*, **71(7):** 2847-2867.

***